Nellai Green & Civic

Civic complaint management with authority email alerts

Report. Notify. Respond. Verify.

N

OUR NELLAI
OUR RESPONSIBILITY

M.E. Computer Science & Engineering
Working localhost demonstration
Tirunelveli / Nellai
Web application, email outbox and community modules

A report should start a response.

Design problem: civic reports and environmental participation need one traceable workflow.

01

Scattered reporting

A concern needs a location, evidence, category and a clear record.

02

Unclear responsibility

Routing should follow configured teams and jurisdictions.

03

Missing closure

A status change should be supported by evidence and citizen feedback.

Project response: a shared, auditable platform that connects reporting with action and community care.

One core platform. Several community outcomes.

- Capture civic and environmental issues with photographs and location.
- Assign responsibility using configurable rules and working-day deadlines.
- Track evidence, escalation and citizen verification.
- Support volunteers, tree records and water observations.
- Automatically queue authority email alerts for every new complaint.

A focused, working localhost demonstration.

A complete complaint workflow with an administrator email outbox; no sensor hardware is required.

Citizen portal

Register, report, track, verify and manage the account.

Authority workspace

Assigned issues, progress updates and resolution evidence.

Administrator controls

Teams, calendars, rules, modules and complaint review.

GIS & environment

Clusters, heatmap, trees and water-resource layers.

Community participation

Volunteer profiles, activities and capacity-aware joining.

Authority email outbox

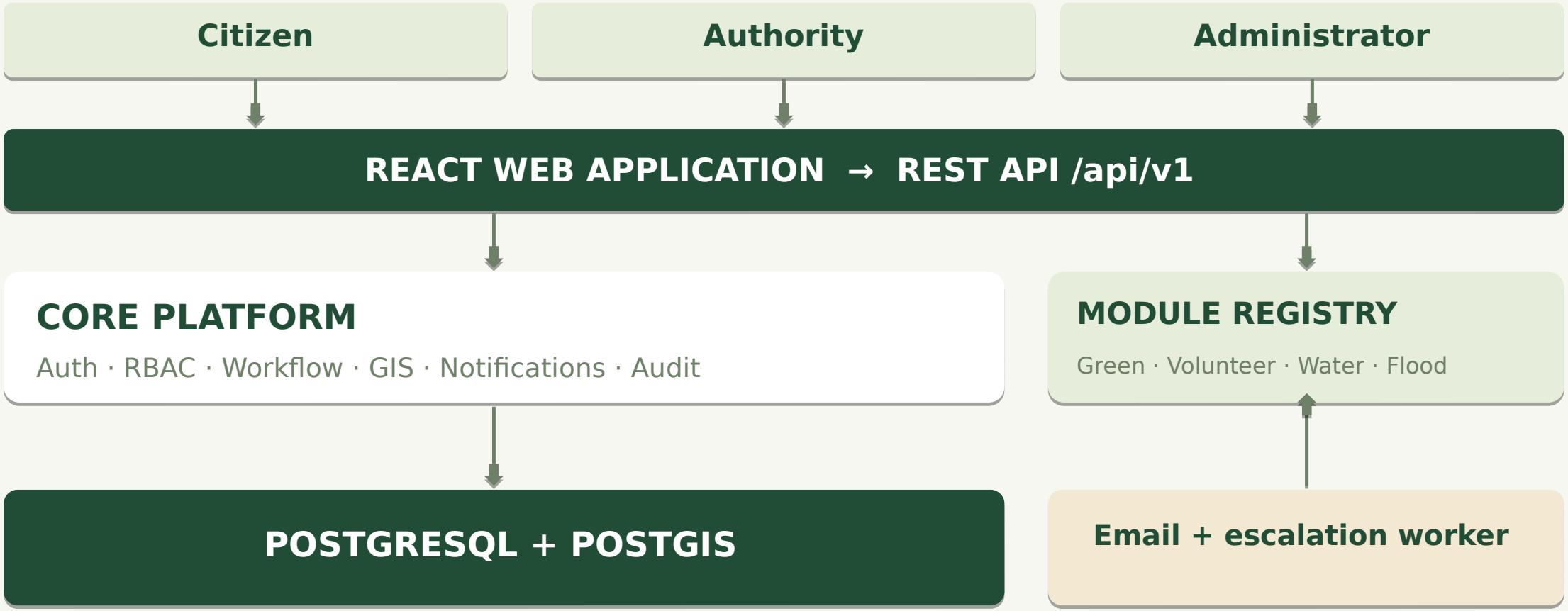
Automatic alerts, message previews and delivery status.

Open technologies, clear responsibilities.

Layer	Technology	Responsibility
Web interface	React + TypeScript	Role-based views, forms and reusable components
API and services	Python + FastAPI	Validation, authentication and complaint workflow
Persistence / GIS	PostgreSQL + PostGIS	Records, distance lookup and geographic boundaries
Maps	Leaflet + OpenStreetMap	Interactive layers, clustering and heatmaps
Development / tests	VS Code, pytest, Playwright	Local run tasks, API and browser verification

SQLite is available as an easier local adapter; PostGIS is used for the prepared demo.

A modular monolith with API boundaries.



Access follows responsibility.

Action	Citizen	Authority	Administrator
Submit a report	Yes	Yes	Yes
View private complaint details	Own reports	Assigned team	Managed reports
Assign team / correct category	No	No	Yes
Update and resolve	No	Assigned team	Yes
Verify resolution	Reporting citizen	Only own reports	Only own reports
Configure modules / calendars	No	No	Yes

Staff role grants require a super administrator. Public maps omit contact details and private photos.

A traceable record behind every action.



Email outbox + audit trail | Optional records: trees · volunteers · water · gauge readings

A complaint ends with citizen verification.

01 Reported → 02 Assigned → 03 Acknowledged → 04 In progress → 05 Resolved → 06 Verification → 07 Closed

Not resolved? Citizen reopens → In progress

Resolution requires a description, photograph, timestamp, location and responsible user.

Deadlines follow a configured working calendar.

Category + GPS + jurisdiction + responsibility rule → responsible team

Example: 3 working days

Reported: Friday, 18 September 2026

Weekend: Saturday and Sunday

Configured example holiday: Monday, 21 September

Deadline: Thursday, 24 September

The holiday is a test-calendar input, not an assertion about official holidays.

Escalation policy

1. Reminder at the due check
2. Configured higher team
3. Additional configured levels

Location informs the response.

01

Nearby search

PostGIS distance queries use metres and spatial indexes.

02

Jurisdiction matching

A point is matched to configured geographic boundaries.

03

Possible duplicates

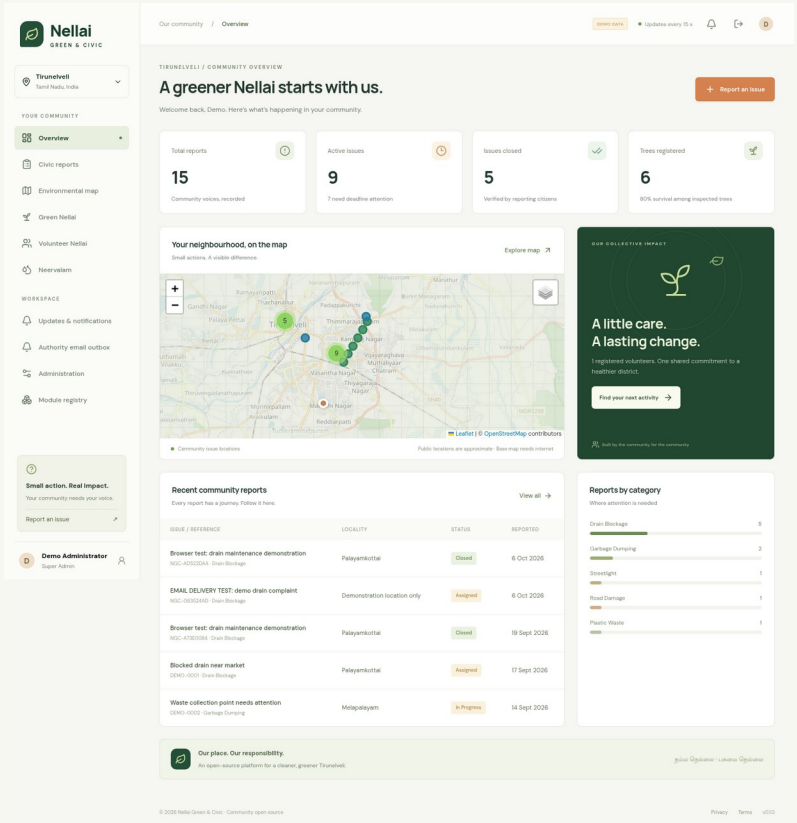
Same category, within 100 m and seven days; text similarity supports review.

Reports are linked after administrator review. Original reports and evidence are retained.

Public map positions are rounded. Official boundaries and ownership need verified records.

The working dashboard.

Actual screenshot from the running local demo; all shown records are demonstration data.



A shared overview

Report counts

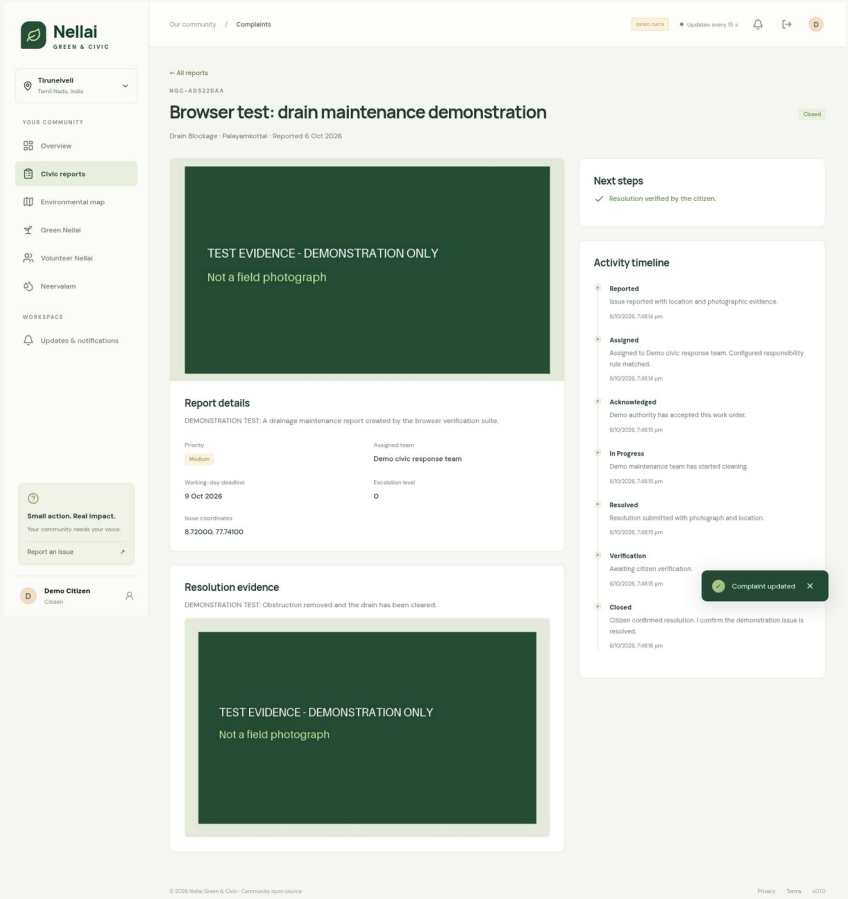
Open and closed issues

Registered trees

Neighbourhood map

Category distribution

Evidence and history stay with the report.



Browser-tested sequence

Citizen uploads a photo

Authority acknowledges

Work is completed

Resolution photo is added

Citizen confirms closure

Every new complaint creates an authority alert.

Complaint submission stays fast because the email is delivered by a separate worker.

1. Save complaint

Report, evidence and an email queue entry are saved together.

2. Preview alert

The administrator sees the recipient, complaint details and status.

3. Deliver or retry

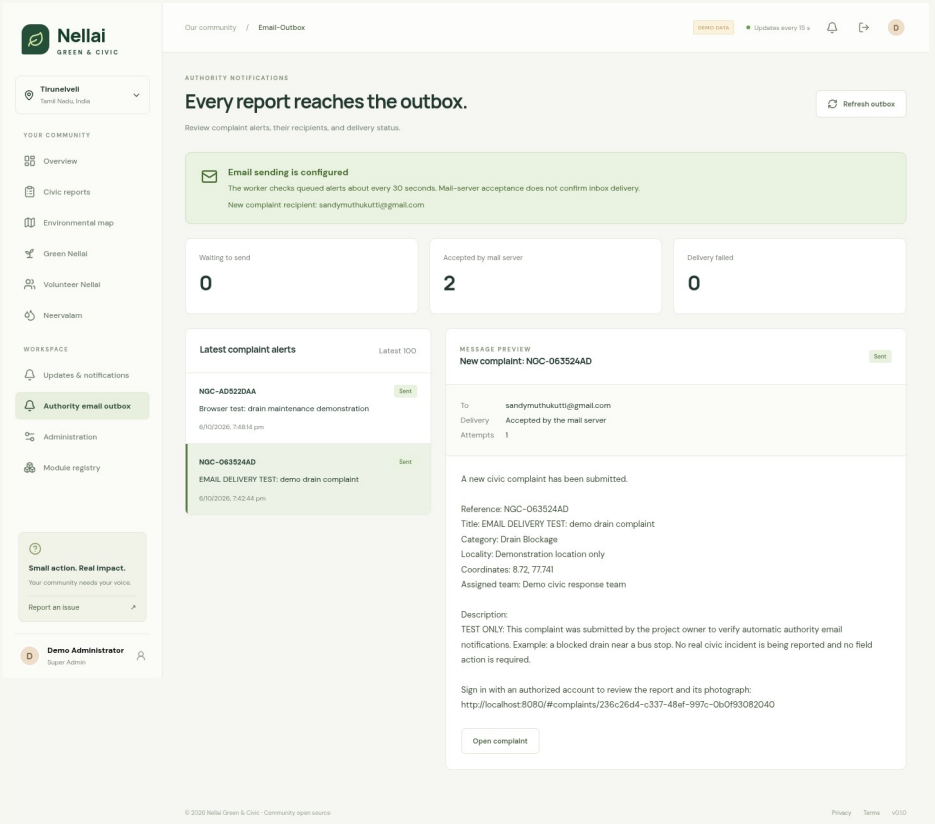
With outgoing mail configured, the worker sends and retries failures.

Recipient for this demo: sandymuthukutti@gmail.com

Verified 7 October 2026: Gmail accepted test NGC-063524AD. Outbox status: SENT.

The authority email, ready to review.

Actual administrator outbox: recipient, message contents and truthful delivery status.



What to demonstrate

Select the test report

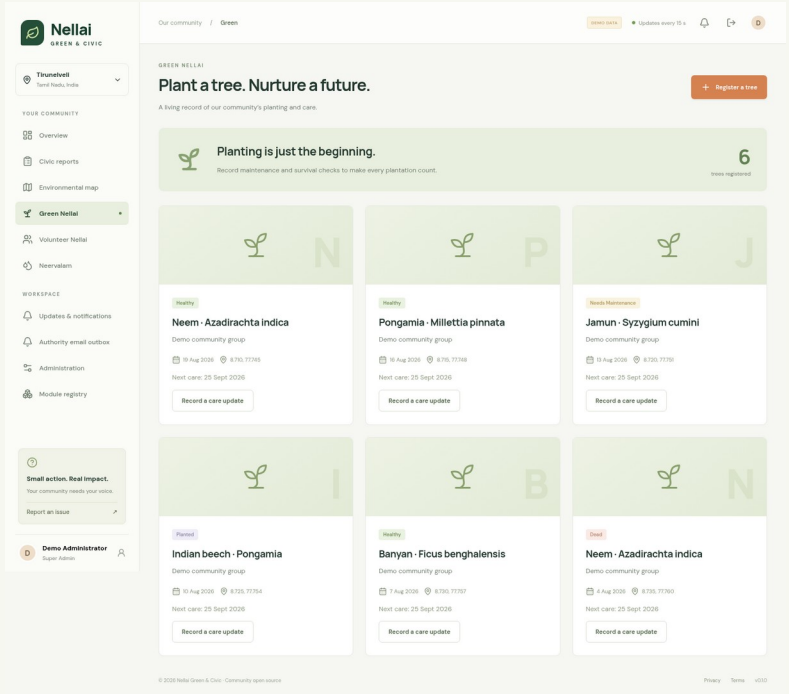
Show the recipient

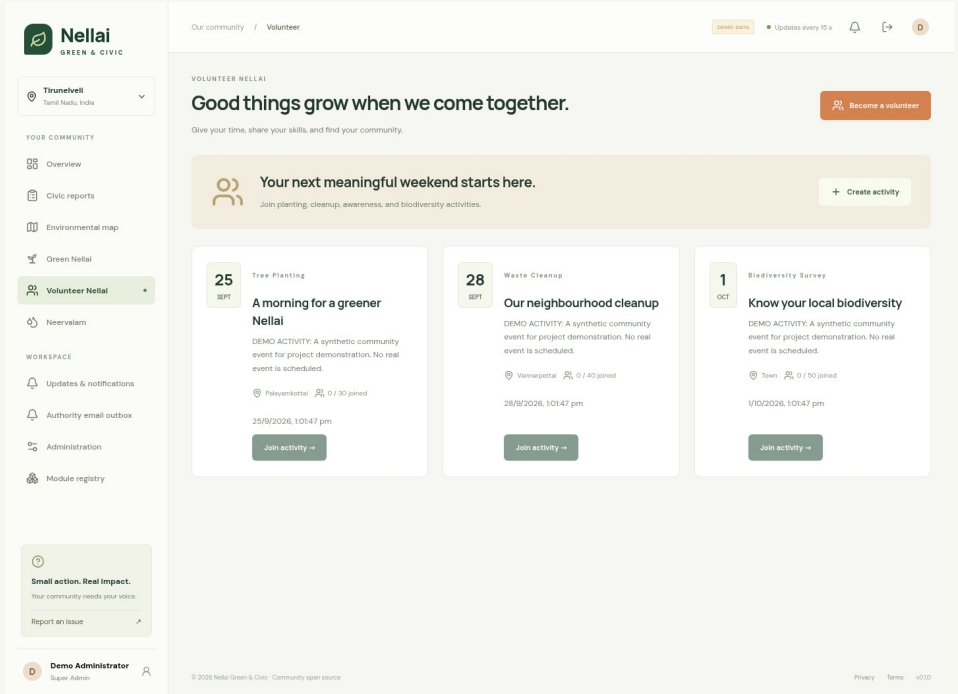
Read the email details

Explain the SENT state

Open its complaint

From a planting day to a living record.

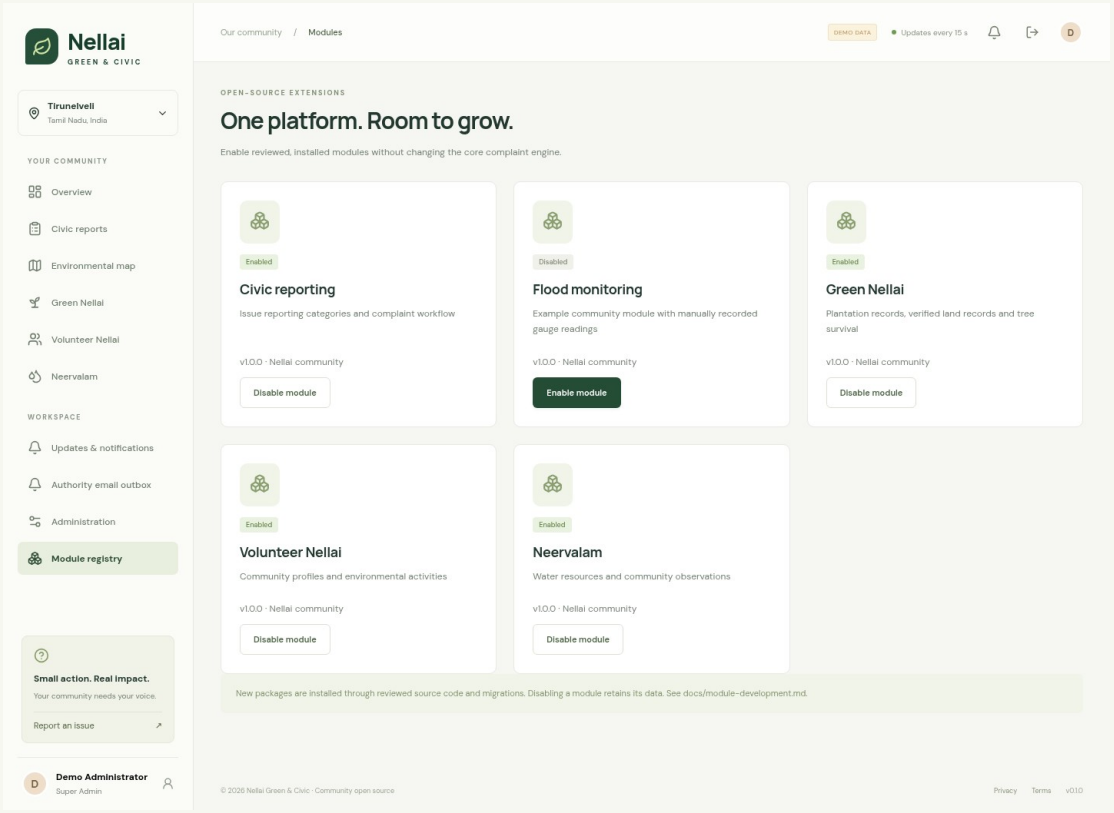




Trees: species, planting date, coordinates and care status.

Volunteers: profiles, activities and available places.

A module can add a new community capability.



MODULE PACKAGE

module.json
backend / API / models
frontend component
database migration
tests + README

Security and privacy are part of the workflow.

Authentication

Argon2 passwords; JWT sessions; refresh rotation.

Authorization

Role and ownership checks on API operations.

Private evidence

Validated images; EXIF removal; authenticated access.

Public-data minimization

No public contact details or private photographs.

Traceability

Workflow history, audit records and session revocation.

User choice

Password-confirmed citizen account deletion.

Verification with reproducible tests.

Results below are functional checks from this implementation, not field-performance research.

25
API / module tests passed
on PostgreSQL + PostGIS

3
Browser scenarios passed
workflow + outbox + layout

PASS
TypeScript + web build
Python lint + migrations

- API checks cover access controls, email retries, pending alerts and private previews.
- Workflow tests cover reopening, duplicate linking and escalation idempotence.

One demo package. Three operating systems.

Extract the ZIP, install Python 3.12, then run the launcher for your PC.

Windows

START_WINDOWS.bat
Double-click to set up
and start the demo.

Linux

bash START_LINUX.sh
Run from the extracted
project folder.

macOS

bash START_MAC.command
Run in Terminal from
the extracted folder.

http://localhost:8080

Built website + SQLite demo data. First setup needs internet; configure Gmail privately on each PC.

A clear boundary between demo and future work.

Working in this demonstration	Further setup or future work
Human-selected complaint categories	Trained image and text classification models
Configured illustrative teams and boundaries	Verified official datasets and government APIs
Manual water observations and gauge records	IoT sensor feeds and validated flood predictions
Responsive React web interface	Completed Flutter app and store publishing
In-app updates and Gmail SMTP alerts	New-PC mail setup; public deployment; mobile push
Functional local tests	Independent security, privacy and performance reviews

Technology references & reproducibility.

FastAPI security and API documentation

https://fastapi.tiangolo.com/tutorial/security/oauth2-jwt/

PostGIS spatial distance queries

https://postgis.net/docs/ST_DWithin.html

React documentation

https://react.dev/learn

Leaflet documentation / OpenStreetMap attribution

https://leafletjs.com/reference.html | https://www.openstreetmap.org/copyright

Source, verification, and demo steps

README.md · docs/verification.md · docs/demo-guide.md

Our Nellai. Our responsibility.

One report. A clear owner. A verifiable resolution.

Nellai Green & Civic

Questions & live demonstration